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CHAPTER 1: INTRODUCTION

1.1. Introduction

Lung cancer is one of the most severe health diseases in the world and is the cause of a lot of deaths each year from the disease. Global health bodies estimate that over 18% of all cancer deaths are due to lung cancer, which is the top form of cancer mortality in both men and women. The high mortality rate is primarily due to late diagnosis of the disease when treatment is less available and less effective. Thus, early and accurate detection of lung cancer has become one of the primary aims of modern healthcare (Esteva et al., 2019).

Medical imaging techniques like chest X-rays and computed tomography (CT) scans are crucial in the diagnosis of lung cancer as they can provide detailed information about the structure of the lungs, helping to identify abnormal regions and determine the stage and presence of a lung cancer tumor. But the images can only be interpreted with the knowledge and experience of radiologists and even minor variations between a harmless nodule and a cancerous one can result in misdiagnosis. Moreover, the volume of medical images has become larger and larger, which puts high demands on medical image analysis, and poses risks for diagnosis errors and delays (Litjens et al., 2017).

Artificial Intelligence has emerged as a crucial component in the healthcare sector, particularly for medical image analysis, in recent years. Deep learning has been able to perform well in challenging tasks like image classification and pattern recognition. Convolutional Neural Networks (CNNs) are very useful because they can automatically learn what is important to the image from the pixels without having to manually select features.

CNNs learn edges, textures and complex image patterns directly from the data, which means manual feature extraction is not required. This has significantly enhanced the accuracy of the automated detection of many diseases, such as lung cancer diagnosis. Transfer learning by using the pre-trained architectures like VGG16, ResNet50, DenseNet etc. is also implemented in the model to enhance its results. These techniques offer higher classification rates and lower training time and computational complexity (Schneider, 2024).

While there has been significant advancement in the use of deep learning to diagnose lung cancer, there are still a number of challenges to its use in practice. Some of the issues pointed out by Adon et al. (2020) are imbalanced data, variable imaging quality among sources, low model interpretability, and low clinical confidence in results of AI systems. The authors state that these problems need to be tackled in systematic ways, before such systems can be used reliably in real healthcare environments.

To address the trade-off between the accuracy of the models and their clinical applicability, Javed et al. (2024) proposed a deep learning system that integrates Convolutional Neural Networks (CNNs) and transfer learning for lung cancer detection. The study also includes Explainable AI techniques to make the model's predictions more transparent, such as Grad-CAM, which provides visual explanations for the model's predictions. This layer can help to establish trust between healthcare professionals and facilitate better clinical decision-making.

1.2. Background of the Study

Lung cancer diagnosis has traditionally been done through medical imaging and pathological examination. Chest X-rays are often used as the initial screening test as they are inexpensive and widely available; CT scans offer more detailed and better quality images, allowing for an accurate diagnosis. These methods have certain drawbacks, however. The detection of lung nodules, particularly in early stages is challenging due to poor image contrast, noise, and overlapping body structures in the images (Eldho et al., 2024).

Traditional image processing techniques (edge detection, thresholding, area segmentation, and morphological operations) were the first applied to the computer-based methods for lung cancer detection. These techniques were used to detect suspicious regions of lungs before moving to machine learning for analysis and classification of the images. These initial automated techniques laid the groundwork for future research, but relied on manually extracted features which required expertise and were prone to errors when applied to other datasets (Pawar et al., 2020).

8 The use of machine learning algorithms like Support Vector Machine (SVM), Random Forest, and k-Nearest Neighbors (k-NN) was a significant improvement in automated lung cancer

detection. These approaches were based on manually extracted features to classify lung nodules. While their classification rates did increase, they did require a lot of feature engineering, making them less suited to larger data sets and new data (Adnan, 2025; Adnan et al., 2025).

10 Deep learning has enabled a new and powerful paradigm to analyze medical images, enabling models to learn directly from raw image data. Convolutional Neural Networks (CNNs) have successfully performed image classification tasks due to their ability to learn spatial relationships among the features in images. These models include various layers such as convolutional layers, pooling layers, and fully connected layers, which are automatically extracting and processing features with different levels of detail in an image (Roy et al., 2025).

To tackle these difficulties, the present project implements sophisticated preprocessing methods like normalization, noise removal and data augmentation, to enhance the quality of the input images and make the model more reliable. Moreover, explainable AI techniques like Grad-CAM can offer visual explanations for the model's predictions, aiding in the understanding of the way choices are made and boosting transparency and trust. Additionally, methods like Grad-CAM, which offer a visual explanation for the model's prediction, can help users understand how the model makes decisions, thereby increasing transparency and trust in the deep learning system..

1

1.3. Research Aim and Objectives

1.3.1. Research Aim

Lung cancer is one of the most prevalent and fatal cancers in the world. Early diagnosis is critical since it will increase the likelihood of treatment success and survival. X-rays and computed tomography scans (CT) are currently used to make the diagnosis, but this is a time-consuming and error-prone procedure that relies on the expertise of the chest radiologist. It is often hard to detect small lung nodules or early cancer signs, particularly when doctors are dealing with numerous medical images. Diagnosis is also difficult due to variations in image quality, patient anatomy and tumor appearance. Current machine learning techniques have proven to be successful, but most require the prior selection of a set of image features that is difficult and labor-intensive. Automatic feature extraction uses Deep Learning, particularly CNN models, and transfer learning models like VGG16, ResNet50, DenseNet and EfficientNet can extract the important features of images and enhance detection accuracy. This study aims to develop a reliable AI-based lung cancer detection system using publicly available medical imaging datasets to improve diagnostic accuracy and reduce human error.

1.3.2. Research Objectives

1. A detailed literature review is to be conducted on lung cancer detection using traditional imaging methods, classical machine learning techniques, and deep learning approaches.
2. Identify, acquire, and preprocess publicly available lung imaging datasets, including proper labeling,
3. A custom CNN model will be developed and trained; including implementation of transfer learning models (VGG16, ResNet50, DenseNet, and EfficientNet) for comparative analysis.
4. All models will be compared using accuracy, precision, recall, F1-score, and ROC-AUC and confusion matrices; with an ethical and practical and clinical perspective for real-world use.

1.4. Structure of Thesis

The thesis is organized into five chapters organized in a logical sequence, starting with the introduction and ending with the conclusions. The Introduction (Chapter One) provides background information on the study, the relevance of enhancing cloud infrastructure performance using adaptive forecasting, problems encountered in the study, objectives of the study, research questions, and the significance of the study. Literature Review is presented in Chapter Two, which focuses on the previous studies on cloud computing, key performance metrics, forecasting methods based on machine learning and research gaps identified in the earlier studies. Research methodology is described in Chapter Three, which covers the research design, data collection methodology, data pre-processing, model development, and model evaluation. The experimental results are presented in Chapter Four, the performance of developed models is analyzed by suitable performance metrics and compared and discussed the results. Finally, Chapter Five concludes with the main findings, contributions to the research, implications for cloud infrastructure and recommendations for future research.

CHAPTER 2: LITERATURE REVIEW

2.1 Literature Review

Checking CT scans by hand to find lung cancer is slow and mistakes can happen easily. Several deep learning models are utilized simultaneously in this work to detect the lung nodule with higher accuracy (not only one model). First, the lung CT images are cleaned up, and each model searches for different features in these images. All the results are put together and the final result is correct about 95% of the time, which means it is a reliable tool for the diagnosis of lung cancer (Shah et al., 2023).

The lung nodules are difficult to detect on computed tomography (CT) due to both the low contrast of the imaging process and the wide range in sizes of the nodules. This approach is based upon a CNN with extra attention to vital locations and features in the lung image. The model is able to distinguish between details and background noise in the image after the cleaning process, outperforming older CNNs in detecting lung nodules that are typically difficult to identify (UrRehman et al., 2024).

Detecting lung cancer early and promptly improves patients' access to treatment and their survival. This method combines machine learning and deep learning for identification and segmentation of lung nodules. It employs 3D CT images of the lungs, a 3D CNN to extract features, Faster R-CNN to detect the nodules and Gradient Boosting to classify them. Following the clean up of images, the model is able to detect 94% of the true nodules and can confidently detect lung cancer in the early stage (Nasrullah et al., 2019).

It can be difficult to identify lung cancer nodules in CT scans due to the diverse appearance of the nodules. This study aims to construct a simple CNN to classify lung images as cancerous or non-cancerous with the help of the LUNA16 dataset. The images get cleaned up after which the model works well, and can be used as a solid basis for an early lung cancer detection model, and later on more advanced ones can be developed as well (Sharma et al., 2018).

It is very important to detect the lung nodules early in the screening process. This paper applies YOLOv5 and CNN feature extraction to locate and classify nodules in the lung CT images. The accuracy is boosted to 92% if the images are cleaned beforehand. This approach is quick and accurate, reducing radiologists' workload when reading lung scans (Thanoon et al., 2023).

Lung nodules are difficult to detect because of their variety of shapes and sizes. In this paper, a multiscale CNN is proposed by extracting features at multi-resolution from the lung images by using Gaussian pyramid. Pre-cleaning the CT scans will make the model detect both small and large lung nodules. Results demonstrate high accuracy, which holds up for the large size variation of nodule imaging in real lungs (Xiong et al., 2025).

The correct diagnosis of whether the lung nodule is harmless or cancerous is important to effective treatment. The aim of this paper is to present a CNN utilizing filters at various scales to get the minute details of lung images. It has been trained with labeled CT data and is able to classify the lung nodules more effectively than conventional CNNs, with potential for automatic lung cancer diagnosis (Ma et al., 2023).

It is crucial to have early detection of nodules in the lungs for effective screening. This paper 1 also uses YOLOv5 with a CNN feature extractor to find and classify lung nodules in CT scans. Preprocessing gives an improvement in time and does not compromise the accuracy (92%). The method is quick, reliable, and reduces the burden on the radiologists interpreting the lung images (Bairagi et al., 2026).

It's important to detect lung nodules early. In this paper, a YOLOv5-based detection and identification system that uses a CNN feature extractor is used for the detection and identification of nodules in lung CT scans. With preprocessing, a performance of 92% accuracy is achieved. It is fast and accurate, and can help to ease the burden on hospital workers who review lung images (Mishra et al., 2019).

Non-cancerous lung nodules are harder to detect when they are accompanied by false alarms. In this paper, two CNN processes are used to enhance the accuracy of this paper. First, a U-Net 1 model splits up the lung CT image to flag possible nodules. Next, a CNN classifier is used to re-check these flagged areas to ensure that they are true lung nodules. The images are pre-processed to remove noise before cleaning. It reduces the number of false alarms without missing true nodules, thus it can be useful in real clinical settings (Cao et al., 2020).

Labeling of lung nodules in CT scan requires a lot of work. In this paper, we suggest a CNN that is trained with global labels without detailed labels. Preprocessing increases the contrast of the image, facilitating detection. When used in the actual clinic, the model is still able to identify lung nodules and roughly outline them, which significantly reduces manual labeling time. (Feng et al., 2017)

In this paper, different CNN models are presented in the classification of lung CT scans into cancer or non-cancer diagnosis aiming to get the best model. First, the quality and consistency of the lung scans are cleaned up, and after the models are trained with a large number of scans. Deeper CNNs are more accurate and provide hints for using certain models to ensure reliable automated diagnosis of lung cancer (Asuntha et al., 2020).

In lung cancer screening, the ability to accurately detect the location of a lung nodule is important, thus the paper applies a region-based lung nodule detection model (Faster R-CNN) to automatically locate lung nodules in lung CT images. It selects candidate regions and classifies them into nodule or non-nodule, while preprocesses the input data to ensure the uniformity. The model is effective at capturing nodules and reducing the number of false alarms, demonstrating that region-based deep learning is an effective approach for lung imaging (Su et al., 2021).

A frequent problem in lung imaging is the limited number of data. This paper applies the transfer learning approach to classify the lung CT scans using pre-trained ResNet and VGG models. The models learn to extract those key features that can distinguish between benign and malignant lung nodules after preprocessing and fine-tuning. Results demonstrate good accuracy, thereby confirming the effectiveness of transfer learning for detecting lung cancer despite the availability of a limited number of data (Sajja et al., 2019).

In this paper, the image cleaning and segmentation technique are used in conjunction with CNN to improve the detection of nodules in lung computed tomography (CT) images. Minimizing noise and maximising contrast emphasizes areas of interest in the lungs. Then, a CNN classifies the cleaned and segmented pictures for locating the cancerous area. Preprocessing is found to improve the performance of CNN in lung imaging by increasing the accuracy and reducing false alarms (Al-Yasriy et al., 2020).

To detect lung cancer early, it's important to make a diagnosis early on. This paper integrates a CNN and an SVM to improve the accuracy: it extracts the features from the lung CT images with CNN, and then classifies the extracted features into nodule or non-nodule with SVM. Pre-processing increases the contrast and normalizes brightness. This is better than CNN alone and is suitable for clinical lung cancer detection applications where high accuracy is required (Murugesan et al., 2022).

Lung cancer detection models should be efficient in extracting various types of features from the CT scan. In this paper, the CNN with multiple layers that is based on feature fusion is constructed to classify lung nodules based on multiple information. The model integrates simple and complex features to enhance the classification of lung images after being cleaned. Lung diagnosis systems are proved to be more accurate and less error prone due to feature fusion (Mahum and Al-Salman, 2023).

It is essential that harmless and cancerous lung nodules be told apart correctly for proper patient care. In this paper, the authors use a CNN to learn from a large lung CT dataset to enhance detection, particularly in challenging cases. Prior to the model inputting the images, the images are normalized and enhanced in the lungs. With a strong classification result, deep learning can help minimize diagnostic errors and aid in better clinical decisions for lung care (Naik and Edla, 2021).

This paper proposes a computer-aided detection (CAD) system based on computer vision with the aim of assisting radiologists in the detection of lung nodules. The system automatically detects the lesions in lung computed tomography (CT) scans with high accuracy and with minimal manual effort. The images of the lungs are then sent to CNN, which decides if nodules are cancerous or not. The system can generate the results with high accuracy, requiring few human checks, and enhancing the clinical workflow for lung cancer diagnosis (Nishio et al., 2018).

This paper uses deep neural networks (DNNs) to track how lung cancer might progress, based on CT scan images of the lungs. The model spots complex patterns in lung nodules and sorts them as cancerous or harmless. The model is enhanced by data augmentation to improve accuracy and robustness. Nishio et al., 2018 shows high accuracy for early detection; deep learning can aid in rapid and accurate lung cancer detection in a clinical environment.

Table 2.1 provides an overview of the strengths, gaps, and future directions for investigations related to lung disease detection, with some challenges highlighted in this field, such as generalization, explainability, and bias, as well as opportunities for enhancing outcomes for deployment.

1 Table 2. 1: Research Gap

Source: Above Literature review.

Ref	**Study Focus**	**Key Strengths**	**Critical Research Gap**	**Proposed Future Direction**
Shah et al., 2023	Ensemble of multiple deep learning models for lung nodule detection	~95% accuracy by combining outputs of several models	Computationally heavy; no discussion of real-time clinical feasibility	Optimize ensemble for lightweight, real-time deployment
UrRehman et al., 2024	Attention-based CNN for nodule detection under low contrast	Better separation of nodule detail from background noise than standard CNNs	Attention mechanism's generalizability across scanners/datasets untested	Validate attention model across multi-vendor CT datasets
Nasrullah et al., 2019	Hybrid 3D CNN + Faster R-CNN + Gradient Boosting pipeline	94% true nodule detection; strong early-stage detection	Multi-stage pipeline adds complexity and potential error propagation	Simplify pipeline into an end-to-end trainable architecture

Sharma et al., 2018	Simple CNN classifier using LUNA16 dataset	Establishes a solid, simple baseline for future models	Limited to basic classification; no advanced architecture tested	Extend baseline with deeper/more advanced architectures
Thanoon et al., 2023	YOLOv5 + CNN feature extraction for nodule localization and classification	92% accuracy; fast, reduces radiologist workload	Accuracy plateau at 92%; unclear performance on small/subtle nodules	Improve small-nodule sensitivity via multiscale YOLO variants
Xiong et al., 2025	Multiscale CNN using Gaussian pyramid for multi-resolution features	Handles wide variation in nodule size effectively	No reported comparison against transformer-based multiscale methods	Benchmark against attention/transformer multiscale models
Ma et al., 2023	Multi-scale filter CNN for benign vs. malignant classification	Captures fine-grained detail better than conventional CNNs	Lacks explainability of learned filters for clinical trust	Incorporate interpretability tools (e.g., Grad-CAM) for clinical validation
Bairagi et al., 2026	YOLOv5 + CNN feature extractor with preprocessing	92% accuracy; time-efficient without accuracy loss	Near-identical approach to prior YOLOv5 studies; limited novelty	Differentiate via domain-specific architecture tuning or new datasets
Mishra et al., 2019	YOLOv5-based detection/classification system	92% accuracy; fast, eases radiologist workload	Repetition of YOLOv5 approach without addressing false positives	Integrate false-positive reduction modules
Cao et al., 2020	Two-stage U-Net (segmentation) + CNN (re-verification)	Reduces false alarms while preserving true nodule detection	Two-stage design increases processing time/resource need	Explore single-stage architectures with false-alarm suppression
Feng et al., 2017	Weakly-supervised CNN using global (non-detailed) labels	Reduces manual labeling effort while retaining rough localization	Coarse localization not precise enough for detailed diagnosis	Combine weak supervision with limited fine-labeled data (semi-supervised)

Asuntha et al., 2020	Comparative study of multiple CNN architectures for cancer classification	Identifies deeper CNNs as more accurate; guides model selection	No unified framework; only comparative analysis, not novel design	Develop optimized deep CNN combining best-performing traits found
Su et al., 2021	Faster R-CNN for region-based nodule localization	Effective nodule capture with reduced false alarms	Region-proposal step may miss atypical nodule shapes	Improve region proposal with shape-adaptive anchors
Sajja et al., 2019	Transfer learning (ResNet, VGG) for limited-data classification	Confirms effectiveness of transfer learning with small datasets	Reliance on pretrained natural-image features may limit CT-specific learning	Explore CT-domain-pretrained backbones
Al-Yasriy et al., 2020	Image cleaning/segmentation + CNN classification	Improved accuracy and reduced false alarms via preprocessing	Preprocessing steps not automated/adaptive to varying scan quality	Automate adaptive preprocessing pipeline
Murugesan et al., 2022	CNN (feature extraction) + SVM (classification) hybrid	Outperforms CNN-alone models; suitable for clinical use	Hybrid pipeline less scalable to large datasets	Test scalability on larger, multi-institutional datasets
Mahum and Al-Salman, 2023	Multi-layer feature-fusion CNN	Combines simple and complex features for improved accuracy	Fusion strategy not compared against attention-based fusion methods	Compare with attention-driven feature fusion approaches
Naik and Edla, 2021	CNN trained on large, normalized/enhanced CT dataset	Strong classification results; minimizes diagnostic errors	No mention of handling class imbalance in large datasets	Address class imbalance and rare nodule types
Nishio et al., 2018	CAD system combining computer vision detection + CNN classification	High accuracy with minimal manual radiologist checks	System not tested across diverse patient demographics/scanner types	Validate CAD system across diverse populations and imaging equipment

CHAPTER 3: RESEARCH METHODOLOGY

3.1. Method of Research

Using medical images, this automated lung cancer detection system has been created in a step-by-step fashion as illustrated in figure 3.1.

The initial stage of the process is data collection. The well-known public datasets are used such as LIDC-IDRI, JSRT and LUNA16 that have CT scans and X-ray images of the lungs for the training and test of the system.

After collecting images they are cleaned up in order to be more useful for the model. This involves normalizing the size of all images, transforming pixel values to be on the same scale, denoising and removing any unwanted markings, and generating extra examples of images by flipping and rotating them. The dataset is then split into three parts: the first half is used to train the model, the second half to check the model's progress as it trains, and the third half for the final test.

The next step is to construct the actual model, which has two approaches. First, CNN is built from scratch to detect vital parts of the lungs and classify the images according to the parts they contain. Second, existing models that have been pre-trained on other large datasets are used and fine-tuned, enabling the system to leverage patterns already captured in other models.

When training these models, there are some techniques that can allow them to learn better and circumvent the risk of memorizing the data rather than learning it. This encompasses optimizers like Adam and SGD, a loss function called cross-entropy, stopping training when it ceases to improve, changing the learning rate throughout training and fine tuning of various parameters to obtain best results.

3 Following training, the accuracy, precision, recall, F1 score, specificity, ROC-AUC and confusion matrix are used to evaluate the system performance: the ability of the model to distinguish between cancerous and non-cancerous images of the lungs.

Finally, one method called Grad-CAM is employed to enable the model's decisions to be more understandable. It shows which regions of the image the model was concentrating on to make its prediction, ensuring that the results are not only accurate but also understandable.

1 Combined, these steps constitute the entire lung cancer detection system (Figure 3.1).

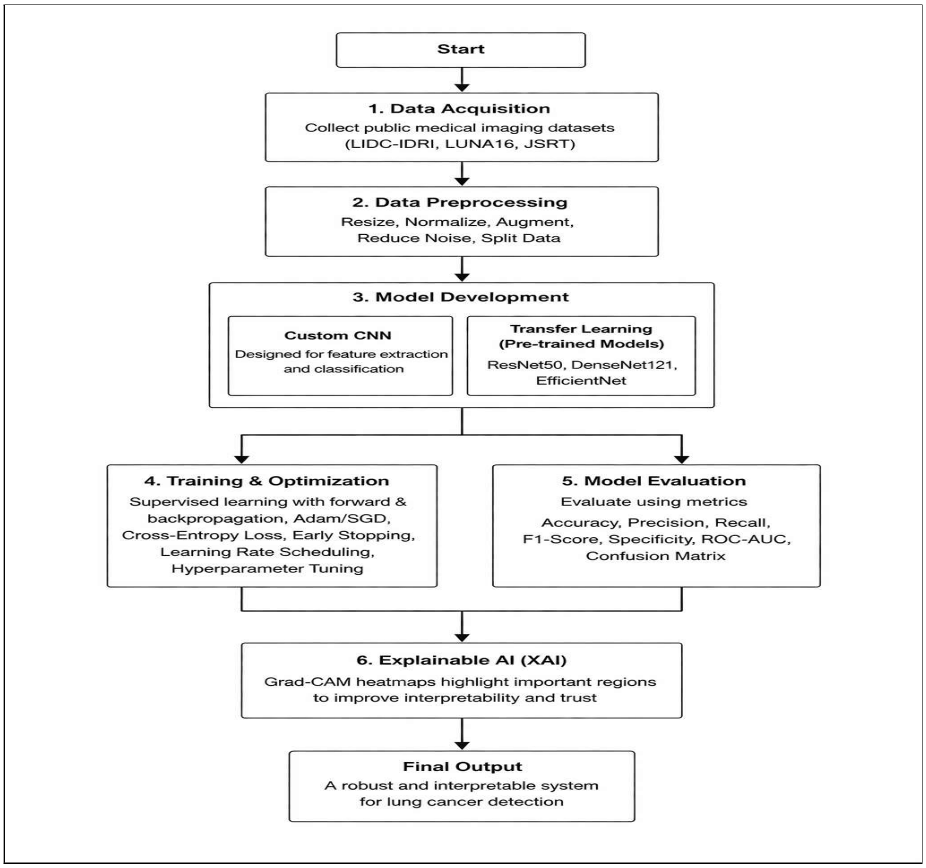

Figure 3. 1: Method of Research

3.2. Data Acquisition

The data collection process is one of the critical factors in this study because the effectiveness of the model depends on the quality and diversity of the data that it is trained with. But it's not that simple to label medical images. Patient information is hard to share due to privacy regulations such as HIPAA, appropriate labelling of each image is a time-consuming process that requires a lot of expertise, and there simply are not enough good datasets available in all locations.

To overcome this, several public datasets were collected rather than one large dataset to better diversify the data, make it more scalable, and allow for future reproduction. The datasets consist of CT scans and chest X-rays from reputable sources such as Kaggle and other open access medical databases. LIDC-IDRI, LUNA16 and JSRT are three datasets that are particularly noteworthy. These are very common in lung cancer research, and are labelled by expert radiologists (Kumari & Goel, 2026).

There is a need for merging these individual datasets into a consolidated dataset, which was applied in the entire study, and this is one of the important contributions of this study. This was done carefully, matching up the labels of each dataset in order to ensure they were classified in the same way (Horry et al., 2023). The final combined data is organized into three groups of lung images:

3 **Normal** – no lung problems are visible

4 **Benign** – nodules not cancerous

5 **Cancerous** – nodules that are malignant

The more data you can combine, the more robust and reliable the model will become: the more data it can be exposed to, the more imaging styles – such as varying resolution, contrast, and scanning methods – it will come across the more robust and reliable it will become. This decreases the chance that the model will only memorize patterns within one set of data, and instead will work well on new and real world lung scans that it has not seen before.

Table 3. 1: Dataset Description

Source: Compiled from publicly available datasets (LIDC-IDRI, LUNA16, JSRT, and Kaggle Chest X-ray dataset)

1

Dataset Name	**Imaging Modality**	**Number of Images**	**Class Distribution (Normal / Benign / Malignant)**
LIDC-IDRI	CT Scan	~1,000+	Malignant dominant (annotated nodules)

LUNA16	CT Scan	~888	Malignant (lung nodules)
JSRT	X-ray	247	Normal and abnormal (mapped to benign/malignant)
Chest X-ray Dataset (Kaggle)	X-ray	~5,000+	Normal / Pneumonia (mapped accordingly)
Custom Unified Dataset	CT + X-ray	~7,000+ (combined)	Normal / Benign / Malignant (balanced after preprocessing)

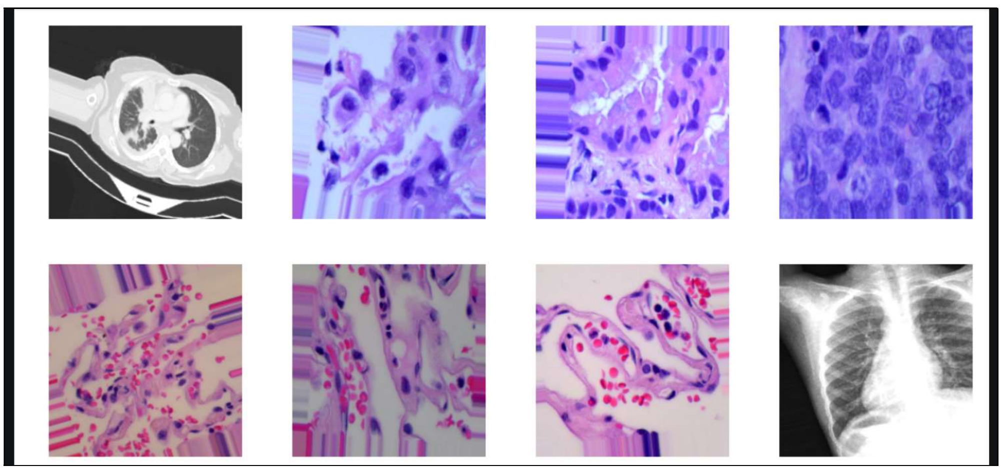

Figure 3. 2: Dataset Images

3.3. Data Preprocessing

A crucial aspect of making medical images ready for the deep learning models to operate on is preparing the data correctly. There are problems, such as noise, inconsistent quality and mismatched resolutions, associated with medical images that, if not addressed first, can adversely affect the model's performance.

The data is extensively preprocessed using a detailed preprocessing pipeline to make it easier for the model to extract useful features from the data. This pipeline involves resizing images to a uniform size, normalizing pixel values, augmenting with extra training examples and noise reduction. All these steps increase data quality, reduce unwanted variation, and ensure suitable format for the deep learning model to correctly use the images (Zhang and Metaxas, 2024).

This preprocessing is not only used to improve the accuracy, but it can also make training faster and improve the model's performance when it is later presented with new lung images it has never seen before.

The model can more easily detect meaningful patterns associated with lung disease like those shown in Figure 3.3 after converting raw images to a clean, standardized format.

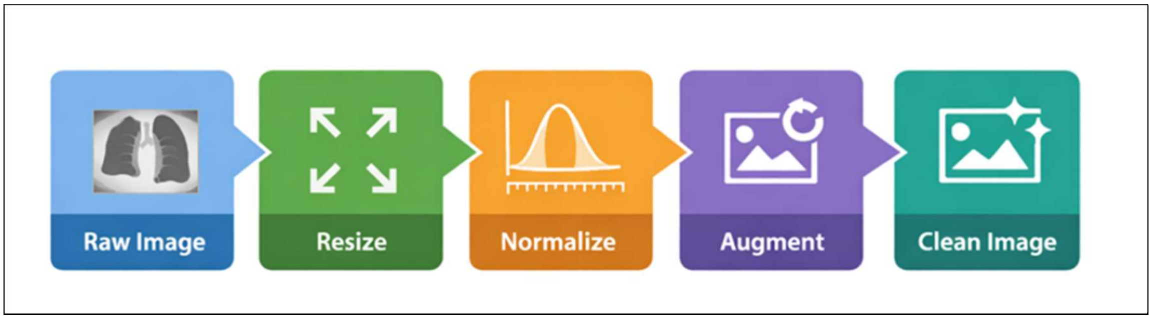

Figure 3. 3: Data preprocessing pipeline

3.4. Image Resizing

The image resizing feature also ensures all the input images are of the same size, a crucial factor in deep learning models. Medical images of various datasets are not homogeneous in their resolution and thus the model input is needed to be standardized. All the images are resized to the fixed size as it is possible to use both custom CNNs and pre-trained CNNs like ResNet50 and DenseNet (Chauhan et al., 2021). This reduces the complexity of the computations and enables a batch to be trained. In addition, resizing also makes the dataset consistent, and the model will not attempt to learn the various sizes of images: it can focus on learning the features of interest.

3.4.1. Normalization

Normalization is used to transform pixel intensity values to a standard range, usually 0 and 1.This is done to make all the input data comparable and it is necessary to have stable and efficient neural network training.

Unless the pixel values are normalized, big changes in pixel values might produce unstable gradients and slow convergence (Kociołek et al., 2020). Normalization standardizes the input data, thus enhancing the learning process and enabling the model to perform more effectively. It also helps to counteract the possibility of numerical instability in training, especially with deep architectures.

3.4.2. Data Augmentation

One of the standard techniques used to generate artificial data that make it more diverse and larger is known as data augmentation. Small datasets and the limited number of examples of malignant cases in comparison to normal ones are especially useful in medical imaging, where they are usually smaller in size and number (Volkova, 2023).

The process is done by using various types of transformations on the images already present, which allows the model to identify them and learn the characteristics that have been transformed. Techniques used here are:

- **Rotate** – to rotate images from slightly different angles
- **Flip horizontally and vertically** – to introduce variation in the data set
- Scaling up and down – to simulate nodules of different sizes
- **Translation** – to move features in the picture to other positions
- **Brightness and contrast adjustment** – to simulate the natural variations in imaging conditions

These techniques can keep the model from overfitting and make it more stable. In contrast to the augmentation, it can be shown many different versions of the same underlying data, thus enhancing its capacity to generalize and function well on novel genuine images it has not been trained on.

3.4.3. Noise Reduction

Medical images may also present noise and unwanted artifacts, which can obscure the clinically important details such as lung nodules. To overcome this, noise reduction techniques are used to enhance the clarity of images and facilitate extracting meaningful features (Goyal et al., 2018).

The filtering technique adopted in this study is Gaussian filtering and Median filtering to remove the unwanted noise and retain the significant structural information of lung images. The model can use this to "clean" the image and concentrate on true patterns associated with lung abnormalities, instead of being misled by the distortion or interference of the image. However, by minimizing noise, the overall quality of the data set is enhanced leading to more accurate predictions by the model.

3.4.4. Dataset Splitting

For an accurate assessment of the model's performance, the dataset was divided into three parts: training, validation, and testing (Ara et al., 2026). The system can learn from one part of the data on the fly and evaluate it with another part of the data, be it during development or during the final evaluation.

The data were split into the following groups:

Training set – 80%

Validation set – 10%

Testing set – 10%

The training set was used to train the model, the testing set for the final performance, and the validation set was used to optimize and tune the parameters of the model, such as the hyperparameter. The structured approach avoids overfitting and guarantees good generalization to new unseen lung images.

The data set was divided with a reasonable balance between the number of malignant and nonmalignant cases, as depicted in Table 3.3 and Figure 3.4. The data set comprises 5,046 training images, 1,044 validation images, and 1,053 testing images, leading to a fair and unbiased assessment of the proposed model.

Table 3. 2: Final Dataset Split and Class Distribution.

Source: Generated from dataset splitting performed in Python during preprocessing (training, validation, and testing sets).

Class Label	**Training Set**	**Validation Set**	**Testing Set**	**Total Images**
Malignant	3,029	626	629	4,284
Normal	2,017	418	424	2,859
Total	**5,046**	**1,044**	**1,053**	**7,143**

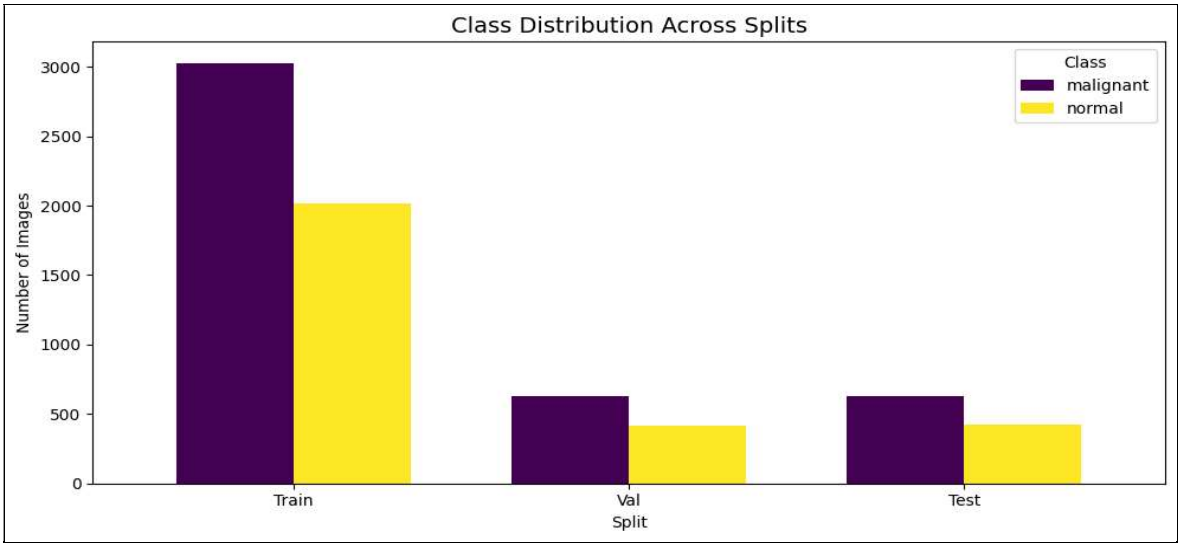

Figure 3. 4: Class Distribution Across Splits

3.5. Model Development

The aim of the model development is to create a sophisticated deep learning model that would be able to classify the lung images correctly into their respective categories. A CNN is used for this purpose, which is designed to learn features automatically from the input images, starting from simple to more complex features (Polat and Danaei Mehr, 2019).

CNNs are created with a number of convolutional layers that initial discover fundamental features, such as edges and textures, and progressively learn more intricate patterns as data passes through the network. These are combined with the pooling layers to compress the data and speed up the computations. To maintain a stable training process and accelerate training, batch normalization is performed, and dropout layers are used to suppress the neurons selected for training randomly, to avoid overfitting.

The final layer in a fully connected network combines all the features extracted by the other layers and classifies the extracted features according to the classification model. In general, the CNN is well balanced in terms of high accuracy and reasonable processing speed, and can provide good classification results on lung images.

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3.6. Transfer Learning

Transfer learning is employed to enhance the model performance of the custom CNN model. It does this by reusing the knowledge that was gained during the training of models on large, general datasets, which saves time and enhances the feature extraction. As part of this study, three of those models—ResNet50, DenseNet121, and EfficientNet — are fine-tuned with lung imaging data to be able to be adapted to this specific task.

Transfer learning is particularly useful in medical imaging where large labeled datasets are often scarce, but is still useful when the amount of data required to produce good results is lower. This enhances the model performance and enable to classify the lung images more efficiently and reliably (Shin et al., 2016).

In summary, transfer learning reduces training time and computational costs, boosts accuracy and enhances the model's ability to perform well on out-of-sample lung images.

3.7. Training Strategy and Optimization

This model is trained in supervised learning by learning from labeled lung images. At training time, the network passes the input through the layers of the network (forward propagation) and then adjusts its internal weights based on the error between the prediction made and the actual target (backpropagation), and the error is decreased over time. The Adam and SGD optimizers are used to classify, and a loss function known as categorical cross-entropy. In order to prevent overfitting, early stopping is implemented to stop the training if the loss does not improve anymore, and learning rate scheduling is employed to ensure the model smoothes out in the training. The parameter values, such as batch size, learning rate and number of training epochs, are also optimized to yield the best results.

3.8. Model Evaluation

After training, a variety of performance metrics are used to assess the performance and reliability of the model, providing a comprehensive view of its classification accuracy on lung images (Alem and Kumar, 2022). These measures include accuracy, precision, recall, F1-score and specificity. The confusion matrix also visually demonstrates the errors made by the model. These results are compared between models, to identify the best model according to accuracy and generalization ability to unseen data.

3.9. Explainable Artificial Intelligence

Explainable AI (XAI) enables better understanding and trust of the model's predictions, particularly in healthcare environments. Grad-CAM is one such tool that generates heatmaps of the areas of a lung image that were most important to the model's prediction. Such visual explanations give doctors confidence in the system and make it more useful to use in real clinic situations, as they can see how the model reached its prediction.

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CHAPTER 4: IMPLEMENTATION

4.1 System Architecture

The proposed lung cancer detection system is a complete deep learning pipeline, which classify CT and X-ray images of the lungs as malignant or non-malignant. It involves collecting data from public datasets, preprocessing it (such as resizing, normalization, and data augmentation), and finally training the model. Features are extracted from a custom CNN and EfficientNetB0, and the predictions from both of these models are combined, or "ensembled" to improve classification accuracy. In order to make easier the decisions made by the system, Grad-CAM is utilized to create heat maps as demonstrated in Figure 4.1.

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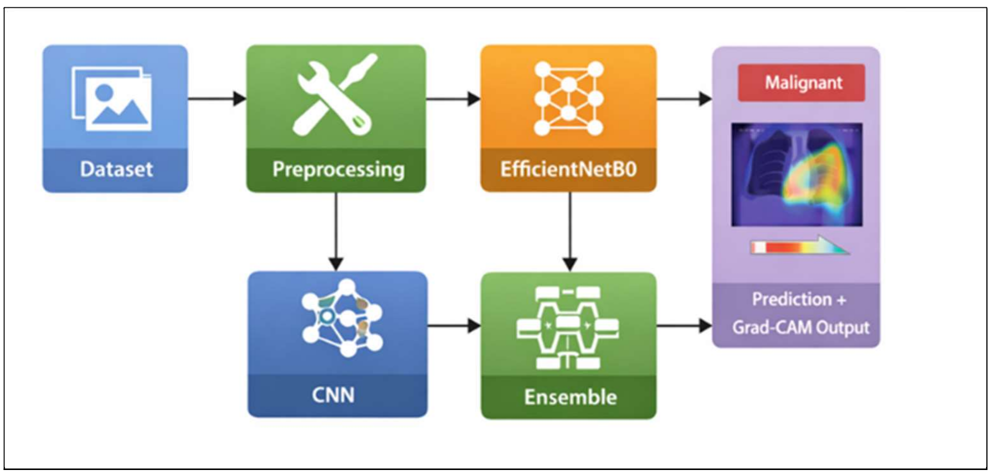

Figure 4. 1: System Architecture

4.2 Model Implementation

The system utilizes two models: a customized CNN and EfficientNetB0, to enhance the ability to accurately classify lung images. The CNN gradually learns an understanding of the image, from a beginning of edge perception to the end of nodule perception, by employing a sequence of convolutional layers, ReLU activation, max-pooling, batch normalization and dropout, and finally, a single yes/no classification fully connected layer. EfficientNetB0, however, leverages a transfer learning approach, where it transfers knowledge already acquired to this specific task. The output of both models are averaged together, a process called ensemble averaging, resulting in a more accurate overall, and more robust system, as shown in Figure 4.2.

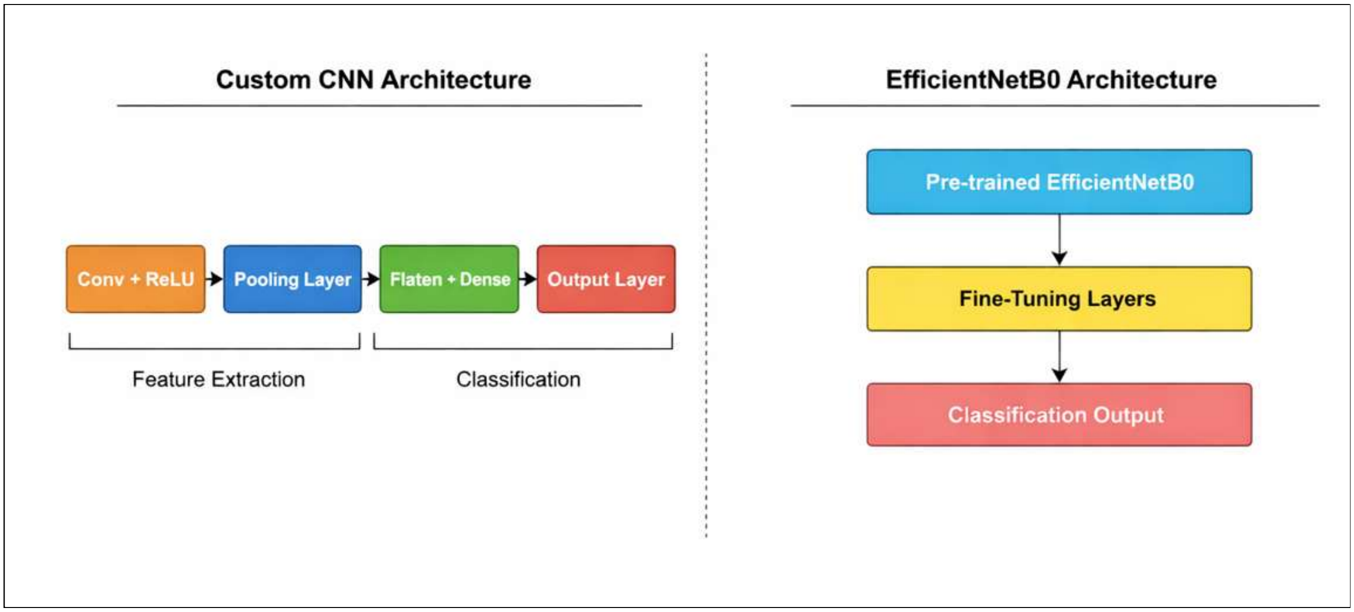

Figure 4. 2: CNN and EfficientNetB0 Architecture Diagram

4.3 Software and Tools

The system is developed and tested by a group of the latest and most commonly used tools. GPU/TPU acceleration was made available by Google Colab, and Python was used for the programming language because it was a flexible option. The deep learning models were developed using TensorFlow and Keras; the data was processed with NumPy and Pandas; and the results were visualized using Matplotlib and Seaborn. As summarized in Table 4.1, OpenCV was employed for image processing while the Kaggle API was employed to manage the datasets to provide a scalable and easily reproducible development setup.

Table 4.1 Software Tools and Technologies used in Lung Cancer Detection system

Source: Based on the tools and technologies used in this project's implementation environment (Google Colab, TensorFlow/Keras, NumPy, Pandas, Matplotlib, Seaborn, OpenCV, Kaggle API, Google Drive), as set up in the project's code environment.

1

Tool / Technology	**Purpose / Role in System**
Google Colab	Cloud-based Python execution with GPU/TPU support for faster training and reduced hardware dependency
Python	Core language for implementing the deep learning system due to its simplicity and rich AI ecosystem
TensorFlow, Keras	TensorFlow handles backend computation and model training; Keras provides a high-level API for designing neural networks efficiently
NumPy, Pandas	NumPy handles numerical computations and matrix operations; Pandas handles structured data manipulation and preprocessing
Matplotlib, Seaborn	Visualizes dataset distributions, training performance, accuracy curves, and loss graphs for model evaluation
OpenCV	Handles image preprocessing tasks such as resizing, filtering, enhancement, and augmentation preparation
Kaggle API	Enables direct downloading and integration of datasets from Kaggle into the development environment
Google Drive	Stores datasets, trained models, and results to ensure persistence and easy access across sessions

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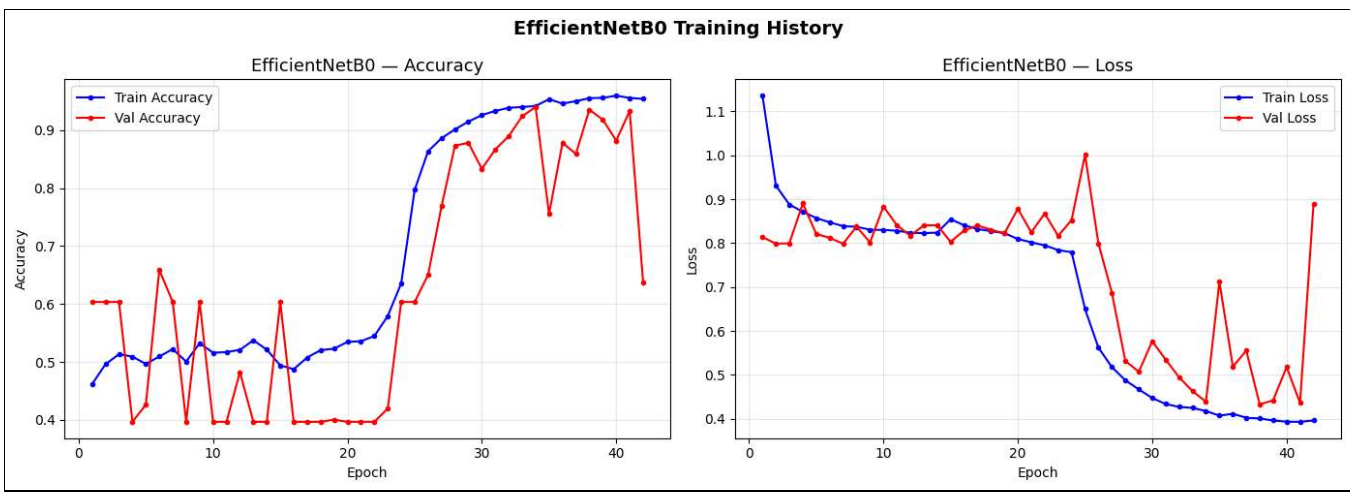
CHAPTER 5: RESULTS AND ANALYSIS

5.1 Experimental Results

This chapter discusses the results of testing three methods for detecting lung cancer: a Custom CNN, EfficientNetB0 and an Ensemble model consisting of both. The challenge is to classify lung images as Normal or Malignant, and the data was divided into training, validation and test sets to ensure that there was no data leakage between the sets and the results were not biased.

Supervised learning was employed to train both models and accuracy, precision, recall, F1 score, and ROC-AUC were used for evaluation. EfficientNetB0 learned faster and more steadily (Figure 5.1) and the Custom CNN had a slower, more gradual learning curve (Figure 5.2). The final predictions and classification results are presented in figure 5.3, which indicates that the system has been working well. Overall, each model works reasonably well, but the ensemble is more accurate and reliable – very important when making medical diagnoses.



1 Figure 5. 1: EfficientNetB0 Training History

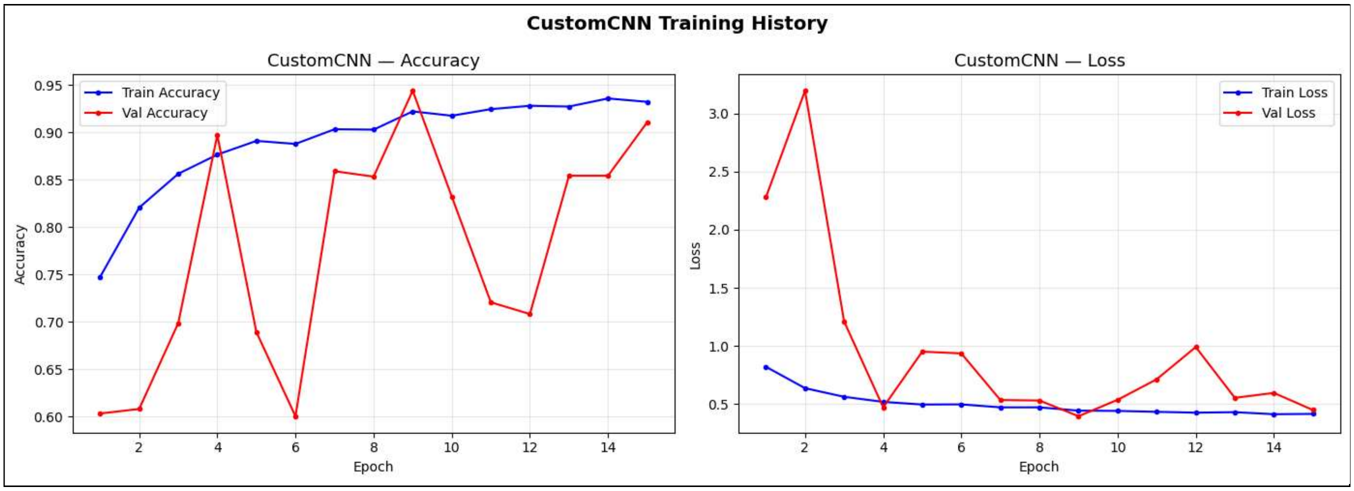


Figure 5. 2 :CNN Training History

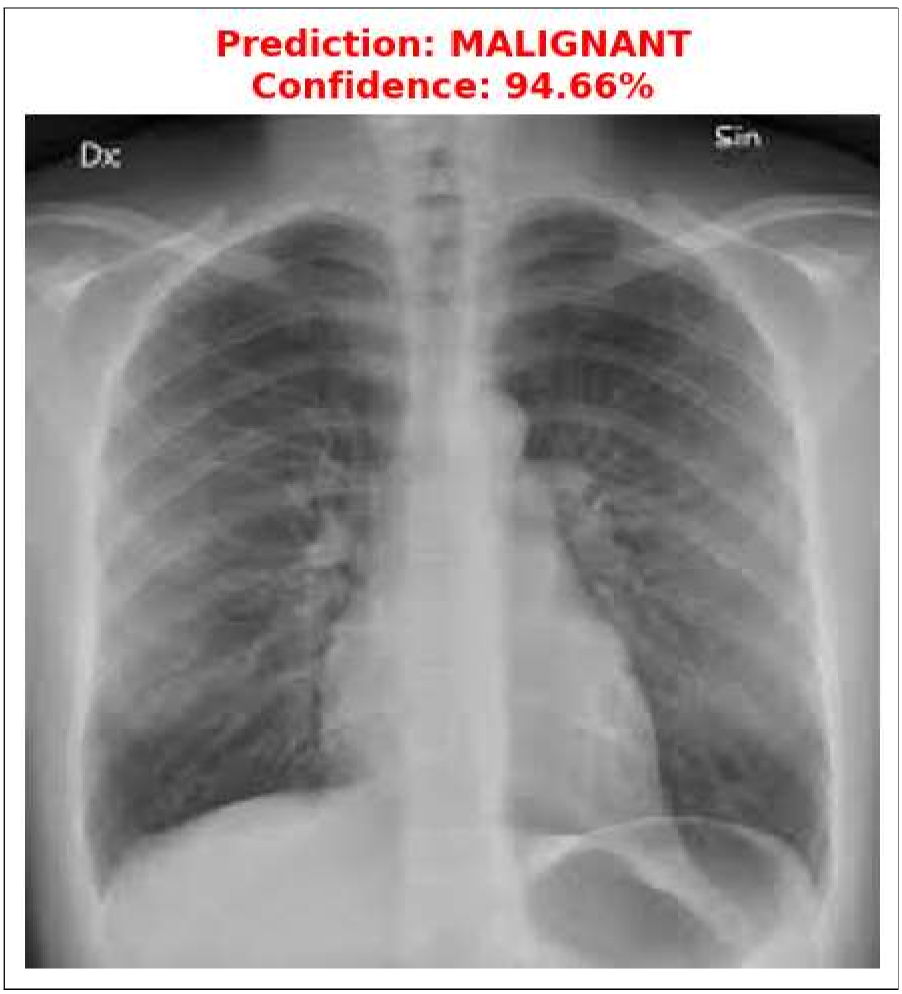


Figure 5. 3: Prediction Result

5.2 Performance

The two main models were compared with respect to their performance and their ability to generalize when tested on new lung images. With less complex patterns, the Custom CNN performs well on spatial features, but less so due to limited training data and no prior knowledge. EfficientNetB0 is trained, however, on the larger ImageNet dataset, which enables it to learn more robust features and generalize better. This advantage allows it to make an accurate classification of the lung images which is demonstrated by the results presented in the Table 5.1.

Table 5. 1: Final Model Comparison (Including Ensemble)

Source: Generated from model evaluation results obtained through Python (TensorFlow/Keras) implementation in this study.

Metric	**CustomCNN**	**EfficientNetB0**	**Ensemble (CustomCNN + EfficientNetB0)**
Accuracy	93.78%	92.58%	**94.45%**
Precision	89.41%	86.27%	**94.0% (macro avg)**
Recall (Sensitivity)	95.43%	96.63%	**95.0% (macro avg)**
F1-Score	92.4%	91.1%	**94.0% (macro avg)**
Malignant Precision	90% approx	86% approx	**98%**
Malignant Recall	94% approx	97% approx	**93%**
Normal Precision	92% approx	90% approx	**90%**

Normal Recall	92% approx	89% approx	**97%**

5.3 Result Analysis

The results of the transfer learning show that the models outperform the models trained from scratch, with EfficientNetB0 being able to learn the complex patterns, as illustrated in Figure 5.5. With data augmentation, the model learns to generalize better and prevents overfitting. With this, the ensemble model performs even better than the CNN alone as seen in Figure 5.4 and supported by the confusion matrix in Figure 5.6.

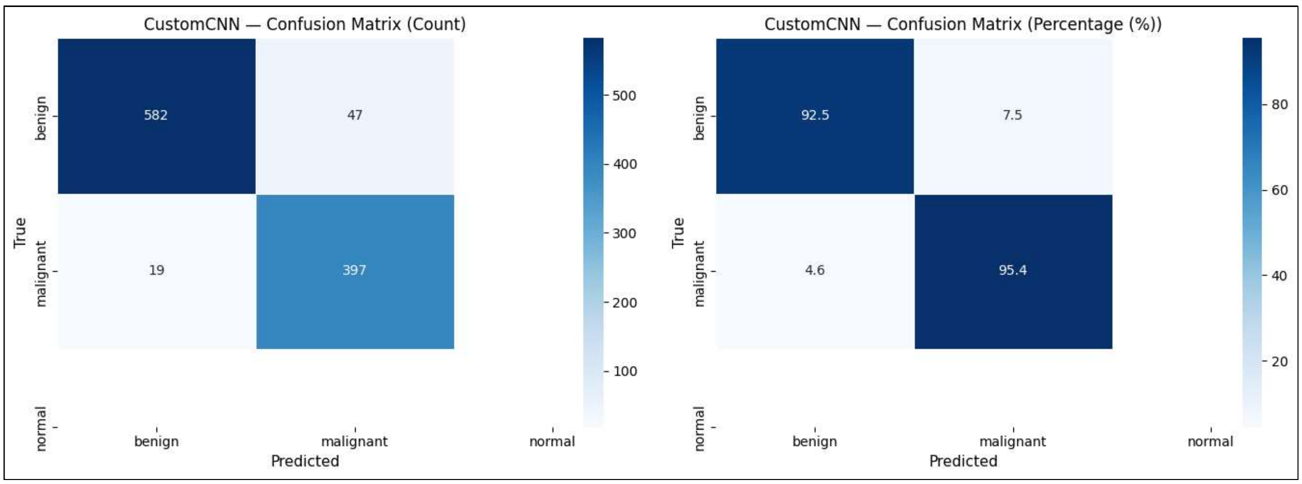


Figure 5. 4: Confusion Matrix Custom CNN

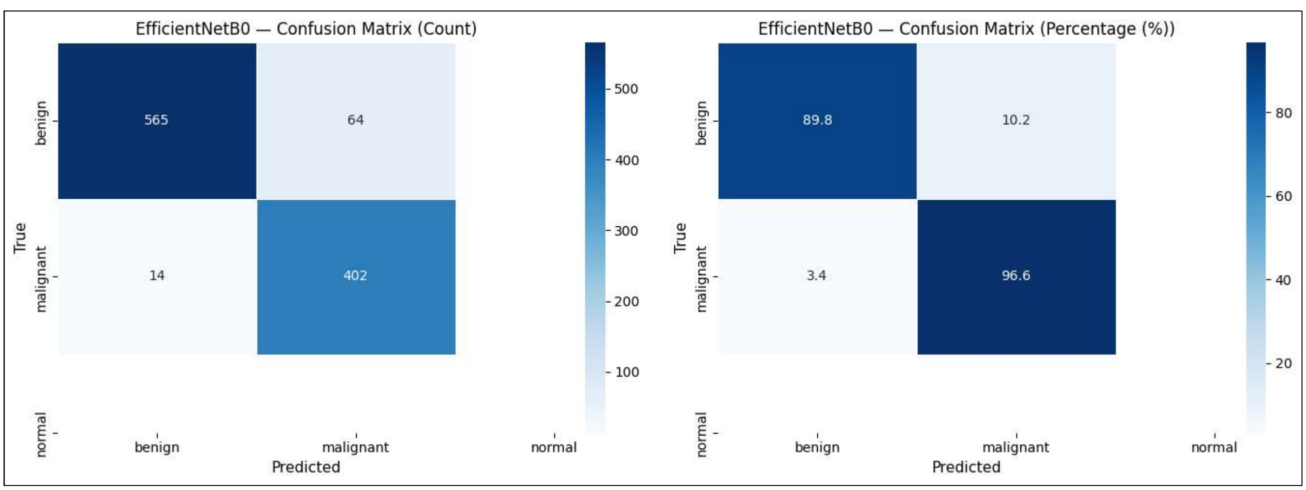


Figure 5. 5: EffcientNetB0 Confusion Matrix

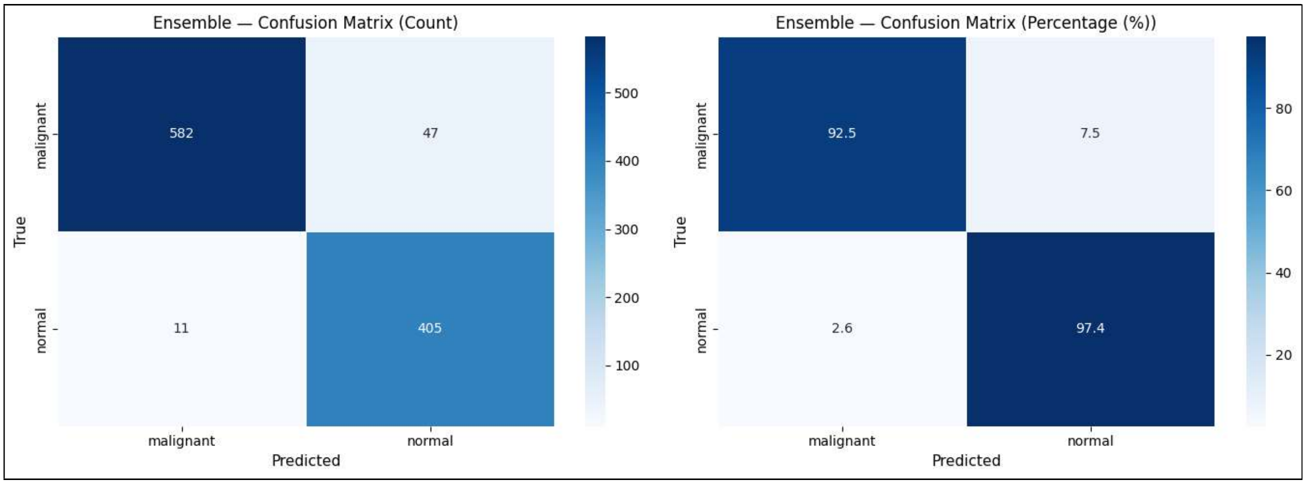


Figure 5. 6: Ensemble Confusion Matrix

5.4 Confusion Matrix Analysis

A confusion matrix is used to categorize the predictions made by a model into true and false positives and true and false negatives, which is important in medical use cases. Overall, most forecasts are accurate with the study demonstrating good accuracy, but a few inaccuracies are seen because normal and malignant cases can appear very similar. A false "no" diagnosis is particularly concerning because it means it is a missed cancer diagnosis. Again, the ensemble model achieves the lowest false negative rate, and a better diagnostic than is possible with either model alone, as per Table 5.2.

Table 5. 2: Model Performance Comparison Table

Source: Generated from model evaluation results obtained through Python (TensorFlow/Keras) implementation in this study.

Metric	**Custom CNN**	**EfficientNetB0**
Accuracy	93.78%	92.58%
Precision	89.41%	86.27%
Recall (Sensitivity)	95.43%	96.63%
F1-Score	92.4%	91.1%
Specificity	92.5%	89.9%

5.5 ROC-AUC Analysis

ROC curve is used for measuring the performance of each model in terms of sensitivity and specificity and AUC is used for an overall performance of a model. The results indicate that both models: EfficientNetB0 and Ensemble model are superior to Custom CNN in terms of AUC. The ensemble model is the best of the three, as illustrated in Figure 5.7.

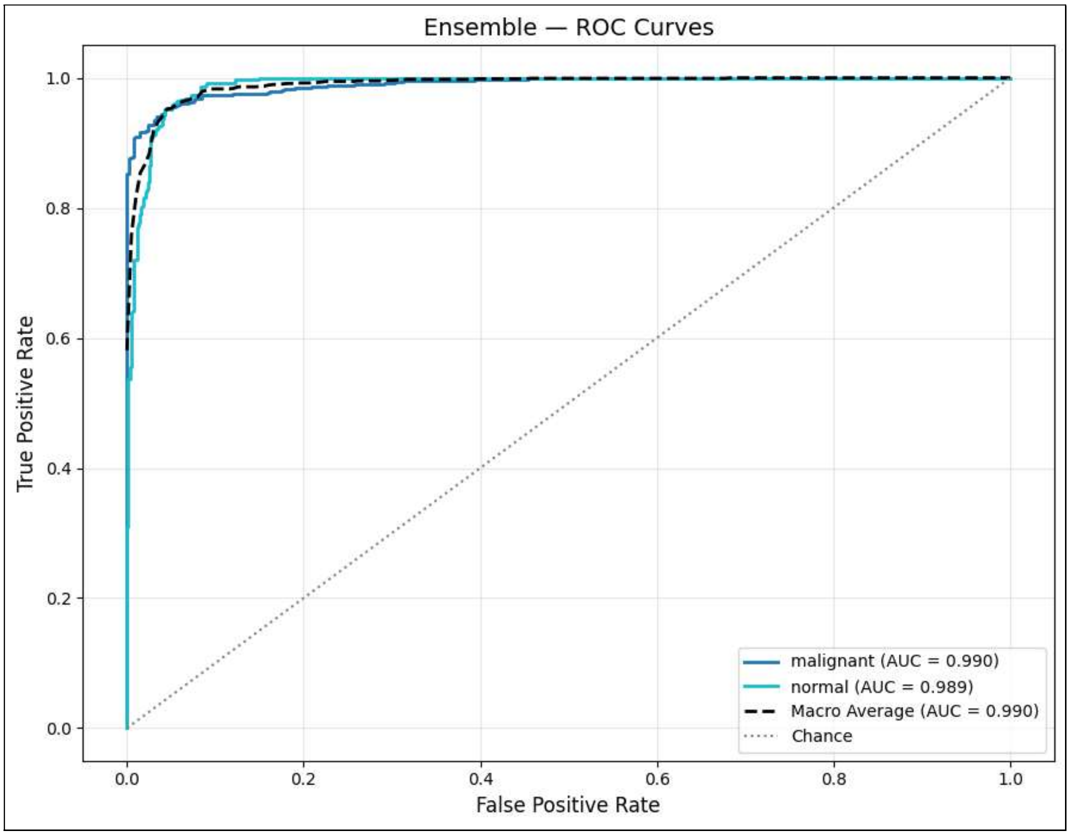


Figure 5. 7: ROC Curve Analysis

5.6 Grad-CAM Visualization Analysis

Grad-CAM visualization helps identify the most important parts of a lung image that affected the model's prediction, which is crucial for instilling trust in healthcare AI. The heatmaps show that the model consistently focuses on the relevant areas of the lung, picking out abnormalities linked to malignant tissue. This type of transparency facilitates trust in the system and creates a more viable environment for clinical usage in lung cancer detection.

5.7 Discussion

Overall, it has been concluded that use of transfer learning and ensemble learning with CNN can outperform the traditional models in detecting lung cancer which are accurate and reliable. The reason is that the transfer learning models have better performance than the standard CNN as they learn the knowledge from large datasets and the ensemble methods enhance the performance by combining the prediction and reducing the error.

However, the quality of the dataset determines the level of performance, and there may be some misclassifications, especially in the case of features that are not well defined or difficult to distinguish in an image. Next, all three models are compared directly in Table 5.3 and the results show that the ensemble model outperforms the other two models overall.

1 Table 5. 3: Performance Comparison of Deep Learning Models (Ensemble, CustomCNN, and EfficientNetB0)

Source: Generated from experimental results obtained through Python-based deep learning models (Custom CNN, EfficientNetB0, and Ensemble) in this study.

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Ensemble	0.944500	0.947400	0.944500	0.944800
CustomCNN	0.936800	0.938800	0.936800	0.937200
EfficientNetB0	0.925400	0.930800	0.925400	0.925900

5.8 Summary

The performance analysis of the proposed Lung Cancer detection system has been presented
1 in this chapter, and it has been concluded that its ensemble model was the best one. Ensemble methods, transfer learning and data augmentation were used to boost the accuracy of classification, and Grad-CAM was used to provide interpretability by identifying the areas of the image that were most relevant to the model's predictions. The next chapter will close the
1 study and provide the suggestions for improvement in the future.

CHAPTER 6: CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project aims to create and evaluate an automatic lung cancer detection system based on deep learning. It was aimed to evaluate the accuracy of the system in detecting lung cancer from chest X-ray and CT scan images with the help of CNN and transfer learning models. The project was divided into a series of phases: data collection, data preprocessing (resizing, image normalization, image noise reduction and augmentation), model construction, and model performance evaluation. Two strategies were employed: ensemble learning using a custom-built CNN and a pre-trained EfficientNetB0 model with transfer learning, enhancing the overall accuracy.

The result indicated that ensemble method achieved the best performance in all metrics: accuracy, precision, recall, F1-score, and ROC-AUC, which could be attributed to the combination of models that mitigated the shortcomings of each individual model and enhanced the generalizability of the model to new lung images. Grad-CAM also provided additional value as it was able to make the model's predictions more explainable by clearly visualizing the portions of the image that affect each prediction. The system in general was accurate, reliable and appropriate to the early diagnosis of lung cancer.

6.2 Future Work

The proposed system could be fine-tuned to enhance its performance, scalability, and suitability for clinical implementation. The use of larger and more varied data sets, particularly from real-world clinical settings, would minimize the impact of biases and enable the model to deliver accurate results across various patient populations and imaging scenarios.

Going forward, additional model architectures could be considered, such as Vision Transformers (ViT) or hybrid CNN-Transformer models, which potentially could yield higher accuracy by recognizing patterns throughout the entire medical image instead of limited local information.

1 For making the system more user-friendly, the techniques such as pruning, quantization and knowledge distillation could be considered to reduce the computational load, making it possible to be deployed on mobile devices or at the edge for real-time diagnosis. The next step would be to integrate it into a comprehensive Clinical Decision Support System (CDSS) to make it more accessible for seamless incorporation into existing healthcare workflows.

Moreover, the integration of explainability features such as SHAP, LIME or attention visualization would help to foster trust in the system's prediction. Lastly, comprehensive clinical evaluation and careful consideration of ethical and regulatory considerations would be crucial before the system could be deployed in clinical practice.

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6.3 Final Remarks

This lung cancer detection system exemplifies the ability of artificial intelligence to offer valuable assistance in medical diagnosis. The system combines the power of deep learning, transfer learning, ensemble learning, and explainable AI, providing a reliable and explainable approach for early detection of lung cancer.

With further optimization and proper clinical validation, systems like this could help doctors reduce diagnostic errors, ultimately contributing to better patient survival rates and a more efficient healthcare system overall.

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